

En (Catherine) Luo

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EDUCATION

The Paideia School | Atlanta, GA, the United States
12th Grade, Class of 2027

Sep. 2023-Current

ACADEMICS

AP

AP English Language and Composition	5
AP Spanish Language and Composition	5
AP Precalculus	5
AP Calculus BC AP Statistics	5
AP Chemistry	5
AP Physics C: Mechanics	5
AP Physics C: Electricity and Magnetism	5
AP Microeconomics	5
AP Macroeconomics	4

SAT

1570 [Math:790 English:780; 99th percentile]

EXTRACURRICULARS

Mathematics

Georgia ARML Team Member (9th–11th Grade):

- Placed top 30 individually in the State Math Tournament each year
- Alternate Team (9th grade), Georgia Team (10th and 11th grade)
- Edited 10+ pages of Power Round proofs

Independent Research (11–12th Grade):

- Literature Review on The Kissing Number Problem in 3D published in The Scholarly Review Journal Fall 2026 Issue
- First Author of Vertex Based Classification of Monohedral Tilings with a Self-Contained Proof accepted and under revision in Journal of High School Science
- An Elementary Family of Multihedral Tilings submitted to Journal of High School Science
- Working paper on Multihedral Tilings planning to submit to RHUMJ
- First author of working paper on Pell Equations planning to submit to Journal of High School Science

Governor's Honors Program - Mathematics Major (11th Grade):

- Conducted research finding and classifying the monohedral, bi-hedral, and tri-hedral tilings of a sphere with rhombi
- Took classes on projective geometry, classical geometry, group theory and math history
- Presented research at 600+ people Majors Fair

Math Team President (9th–12th Grade):

- Initiated 1 new Varsity & JV Team
- Coached weekly practices for 40+ people
- Created a website compiling notes and resources tailored for the school math classes and math team training (www.pimathteam.org)

Math Lunch Tutoring (11th Grade):

- Founded lunch tutoring sessions
- Presented to 400+ students
- Helped 50+ struggling students at 5+ classes of math in 3 months

Music & Art

Atlanta Young Singers (AYS) (9th–12th Grade):

- 1st Soprano
- Performed in Christmas with the Atlanta Symphony Orchestra for 1500+ audiences (Handel's Messiah, Ceremony of Carols)
- Soloist (Biebl's Ave Maria)

Concert Band (9th–12th Grade):

- First Chair Flute and piccolo
- Woodwinds Section Leader
- Performed for 500+ audiences
- Arranged music for small ensemble performances

Paideia School Musical Pit Orchestra Conductor and Player (10th–12th Grade):

- Conductor of each show and rehearsal for Little Shop of Horrors (12th)
- Arranged parts for different instruments; wrote string (violin, viola & cello) and reed (bassoon & trombone) parts
- Worked individually with each player or section on their part
- Coordinated communication with cast, tech and pit
- Principal Flute, Piccolo, (Alto Flute) & Woodwinds section leader (10-12th)

C&H Art Studio (10th-12th Grade):

- Studied under SCAD professor Chin-Cheng Hung
- Showcased tricolor portrait at C&H Open Studio Exhibition

Community Service**Immigration Action Society (11th - 12th grade)**

- Club co-founder
- Collaborated with leaders in planning 5+ meetings and club activities within 3 months
- Coordinated a group research project on underrepresented issues affecting immigrant communities
- Active participant and organizer in walkouts and protests

STM Church Immigrant Support Program(10th grade):

- Volunteers to visits detainees at Stewart detention center and supports them on deportation court cases
- Supports immigrant families with El Refugio
- Translates in Spanish for the group during visits to the detention center and communicates with immigrant families

Focolare Movement (9th-12th grade):

- Organizes monthly youth meetings and immigrant support events
- Plays flute at services

Athletics**Varsity Women's Basketball Team (9th–10th Grade):**

- Shooting Guard
- Helped team qualify for state
- Active for 150+ hours per season

HONORS**Math**

- CEMC Euclid Distinguished Honor Roll (Top 1%) - 87/100
- Berkeley Math Tournament Online 2025: Honor Roll
- CEMC Hypatia Distinguished Honor Roll (Top 1%) - 32/40
- KSU Math Competition: Ranked #17 out of 1300+ students

Music

- GMEA All State 11/12 Symphonic Band 5th chair 2026
- District Honor Band 2025, 2026

Art

- Scholastic Art & Writing Awards 2025: 1 Regional Gold Key + 1 Regional Silver Key in Drawing
- Celebrating Art Contest 2025, 2026: High Merit Award
- Art Renewal Center 18th Salon Competition: National Finalist

RESEARCH & PUBLICATIONS

Vertex-Based Classification of Monohedral Spherical Tilings with Rhombi *Jun. 2026-Jul. 2026* **Independent Researcher**

- First-author of paper exploring and classifying the entirety of configurations of monohedral tilings of a sphere with rhombi
- Accepted and under revision in the Journal of High School Science

A Survey on the Kissing Number in 3D *Feb. 2026-Apr. 2026* **Independent Researcher**

- Reviewed and compared solution methods to the Kissing Number Problem in 3D
- Published on The Scholarly Review Journal Fall 2026 Issue
- <https://doi.org/10.70121/001c.169632>

An Elementary Family of Dihedral Spherical Tiling with Rhombi *Jul. 2026-Sep. 2026* **Independent Researcher**

- Submitted to the Journal of High School Science

Vertex-Based Classification of Multihedral Spherical Tilings with Rhombi *Jun. 2026-Current* **Independent Researcher**

- Explored configurations of monohedral, bi-hedral, and tri-hedral tilings of a sphere with rhombi, classified 1000+ cases
- Planning submission to RHUMJ

A Proof for Solving Pell Equations using k-Fibonacci Sequences *Jun. 2026-Current* **Independent Researcher**

- Explored the relationship between k-Fibonacci sequences with a special form of Pell Equations, proved the solution using the irrational roots method with high-school algebra
- Planning submission to The Journal of High School Science

SKILLS

- Chinese (native), English (proficient), Spanish (Intermediate High)
- Web development (Squarespace, Webflow), Typesetting (LaTeX)
- Red Cross Certified Standard CPR & First Aid